Stage-Resolved Transcriptomic Analysis Reveals Distinct Molecular Programs Associated with Liver Fibrosis Severity

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Abstract

Background: Fibrosis stage is the strongest histological determinant of outcome in metabolic dysfunction-associated steatotic liver disease (MASLD), yet the hepatic transcriptional changes that accompany each step from F0 to F4, and their reproducibility across cohorts, remain incompletely characterised. Most transcriptomic studies compare broad disease categories or pool fibrosis stages, which obscures when individual molecular programmes change.

Methods: We integrated eight public human liver RNA-sequencing datasets (1,085 samples; 1,027 recorded patients) at the patient level and linked them to the OptimusKG biomedical knowledge graph (126,999 nodes; 35,160,254 relationships). Within-study Spearman correlations with fibrosis stage (610 patients) and ordered control–NAFL–NASH labels (216 patients) were combined across studies with a requirement for directional agreement. Fibrosis-associated genes were clustered by their F0–F4 profiles, adjacent stages were compared with study-adjusted models and size-matched refits, and genes were prioritised with a nine-point multi-evidence score assessed by leave-one-study-out (LOSO) and sensitivity analyses.

Results: We identified 3,845 fibrosis-associated and 4,199 disease-group-associated genes, 1,822 of them shared with strongly concordant effects (Spearman $\rho = 0.702$). Fibrosis-associated genes followed five distinct expression patterns across fibrosis stages: higher expression from F1 that increased further at each later stage, including fibrillar collagens, *PDGFRB* and *TGFB1*; higher expression from F1 without further increase, enriched for E2F targets and proliferation markers; an increase largest between F3 and F4, enriched for epithelial–mesenchymal transition, basement-membrane remodelling and ductular-reaction markers; a decrease largest between F1 and F2; and a decrease largest between F3 and F4 of bile-acid, fatty-acid and drug metabolism genes. The patterns nested within the three-cluster solution (91.9% concordance) and agreed with study-adjusted stage models, and at equal sample size only the F3–F4 transition showed many substantial differences, although F4 was represented by only 34 patients. Prioritisation identified 447 Tier 1 genes and a 23-gene shortlist spanning all patterns, including *PDGFRA*, *TREM2* and *CACNA1C*; 442 Tier 1 genes remained significant in every LOSO refit, and knowledge-graph analysis linked them to matrix, adhesion, integrin and PDGF signalling annotations.

Conclusions: Liver fibrosis severity is associated with distinct stage-dependent transcriptional programmes rather than a single monotonic expression response, with fibrogenic matrix genes already elevated at mild stages and ductular, remodelling and metabolic differences most pronounced between F3 and F4, where estimates remain least precise. The integrated, uncertainty-aware resource provides prioritised, stage-annotated candidates for cell-resolved and functional validation.

Keywords: Liver fibrosis; MASLD; Transcriptomics; Meta-analysis; Gene expression patterns; Knowledge

graph; Gene prioritisation; Fibrosis stage

Introduction

Fibrosis is the central consequence of chronic liver injury and the histological feature most strongly associated with long-term outcomes in fatty liver disease [1, 2]. In a prospective cohort of adults with non-alcoholic fatty liver disease (NAFLD), the risks of liver-related events and death increased with each fibrosis stage [3]. The disease, now termed metabolic dysfunction-associated steatotic liver disease (MASLD) and steatohepatitis (MASH) [4], affects an estimated 30% of adults worldwide [5], and fibrosis is staged histologically from F0 to F4 [6]. Progression is nevertheless heterogeneous: in paired biopsies, fibrosis may advance, remain stable or regress [7], and effective antifibrotic therapy remains a major unmet need [8]. Understanding which molecular changes accompany each step in severity is therefore important both for explaining this heterogeneity and for identifying stage-appropriate biomarkers and targets.

Liver transcriptomics has repeatedly linked gene expression to histological severity. Early microarray work separated mild (F0–F1) from severe (F3–F4) NAFLD fibrosis [9]. Hoang *et al.* analysed 78 biopsies across the severity spectrum and identified progressively regulated genes [10]; Govaere *et al.* profiled 206 patients, derived a 25-gene signature of fibrosing steatohepatitis and evaluated it in an independent cohort [11]; and other cohorts have characterised fibrosis together with cell-type composition [12] or identified circulating biomarkers such as thrombospondin-2 [13, 14]. Because single cohorts are limited in size and stage coverage, several studies have integrated public data: meta-analyses of NAFLD and MASLD progression [15, 16], comparisons of fibrotic and non-fibrotic liver across aetiologies [17], and a multi-cohort analysis of about 1,000 patients identifying 846 genes that differ between advanced (F3–F4) and mild (F0–F1) fibrosis [18].

These studies establish that reproducible fibrosis-associated genes exist, but they summarise progression as a contrast between categories (Table 1). Pooled designs cannot show whether a gene already differs at mild stages, at intermediate stages or only between F3 and F4, although this stage pattern determines whether a gene is a plausible early biomarker, a marker of advanced disease or a target whose relevance depends on stage. Resolving it requires several methodological safeguards: harmonising expression units, gene identifiers and clinical labels across cohorts; handling repeated biopsies so that some patients do not carry extra weight; correcting for unequal stage sizes; and testing whether derived patterns are robust to the choices used to obtain them [19, 20, 21]. Biomedical knowledge graphs provide a complementary layer for interpreting the resulting genes by linking them to pathways, diseases and drugs [22, 23, 24, 25], but are rarely combined with patient-level, stage-resolved expression evidence.

Here we integrated eight public human liver transcriptomic datasets at the patient level with the OptimusKG knowledge graph (Fig. 1). We asked (i) which genes are associated, consistently across studies, with fibrosis stage and with ordered disease-group labels; (ii) which expression patterns characterise the progression from F0 to F4, and how robust these patterns are; and (iii) which genes are best supported when association strength, stage pattern, transition evidence and functional annotation are combined. The principal contributions are an integrated transcriptomic and knowledge-graph resource; a cross-study map of fibrosis- and disease-group-associated genes resolved into five stage patterns with explicit robustness assessment; a transparent multi-evidence prioritisation framework; and a biological interpretation linking each pattern to the cell types and processes of the fibrotic liver.

Table 1. Reported features of representative liver transcriptomic studies and the present study. ✓ reported; ✗ not reported for the defined feature (not proof that it was never done). Gene stage-profile clustering means clustering genes by their F0–F4 profiles, not clustering samples.

Study	Multiple cohorts	Separate F0–F4	Gene stage-profile clusters	Adjacent-stage contrasts	Knowledge graph	External/exp. follow-up
Hoang *et al.*, 2019 [10]	✗	✓	✗	✗	✗	✗
Govaere *et al.*, 2020 [11]	✓	✓	✗	✗	✗	✓
Piras & DiStefano, 2024 [16]	✓	✗	✗	✗	✗	✓
Yang *et al.*, 2026 [17]	✓	✗	✗	✗	✗	✓
Maiers *et al.*, 2026 [18]	✓	✗	✗	✗	✗	✓
This study	✓	✓	✓	✓	✓	✗

Methods

Study design and data sources

We conducted a retrospective computational study of human liver transcriptomic data deposited in the NCBI Gene Expression Omnibus (GEO) [26]. Eight datasets were analysed: GSE126848 [27], GSE130970 [10], GSE135251 [11], GSE162694 [12], GSE167523 [13], GSE193066 [28], GSE240729 [14] and GSE269412 [29] (Table 2). Datasets were identified by a targeted search of GEO for human liver transcriptomic studies of NAFLD/MASLD with histological fibrosis staging or disease-group annotation. Inclusion criteria were: (i) human liver tissue; (ii) bulk RNA-sequencing data deposited as sample-level expression files; (iii) metadata linking each measurement to a sample and, where available, to a patient and biopsy; and (iv) for an association analysis, fibrosis stage (F0–F4) or ordered control/NAFL/NASH labels for at least 25 patients with variation on that clinical axis. Series of non-human, cell-line, organoid or single-cell data, non-liver tissue, or without linkable sample-level expression were excluded. Datasets meeting criteria (i)–(iii) but lacking clinical annotation (GSE269412) were retained only in the knowledge graph. Because the search was targeted rather than a formal systematic review, the cohorts are not claimed to be exhaustive. Deposited expression matrices (gene counts, TPM, per-sample files or GCT; Additional file 1: Table S5) and series-matrix metadata were used without realignment or requantification. No new human data were collected.

Clinical annotation and patient handling

Expression samples were linked to GEO records using study-specific identifier, title and description fields, and clinical information was harmonised into a common structure (study, sample, patient, biopsy order, fibrosis stage, disease group) with original fields and provenance retained. Fibrosis severity was represented by ordinal stages F0–F4 [6]. Disease groups were ordered as control (0), non-alcoholic fatty liver (NAFL; 1) and non-alcoholic steatohepatitis (NASH; 2). Broad NAFLD labels without an informative subtype, obese-only labels and unclassified records were excluded from the disease-group analysis, and original diagnoses were not assumed equivalent to contemporary MASLD/MASH criteria [4]. For GSE130970, individual author-assigned diagnoses were not available; disease groups were therefore derived from deposited histological scores (steatosis grade 0, control; steatosis with hepatocyte ballooning, NASH; steatosis without ballooning, NAFL) [10]. Five ambiguous samples (GSM3758028, GSM3758038, GSM3758039, GSM3758073 and GSM3758074) were set to unclassified for this axis but retained their fibrosis stages.

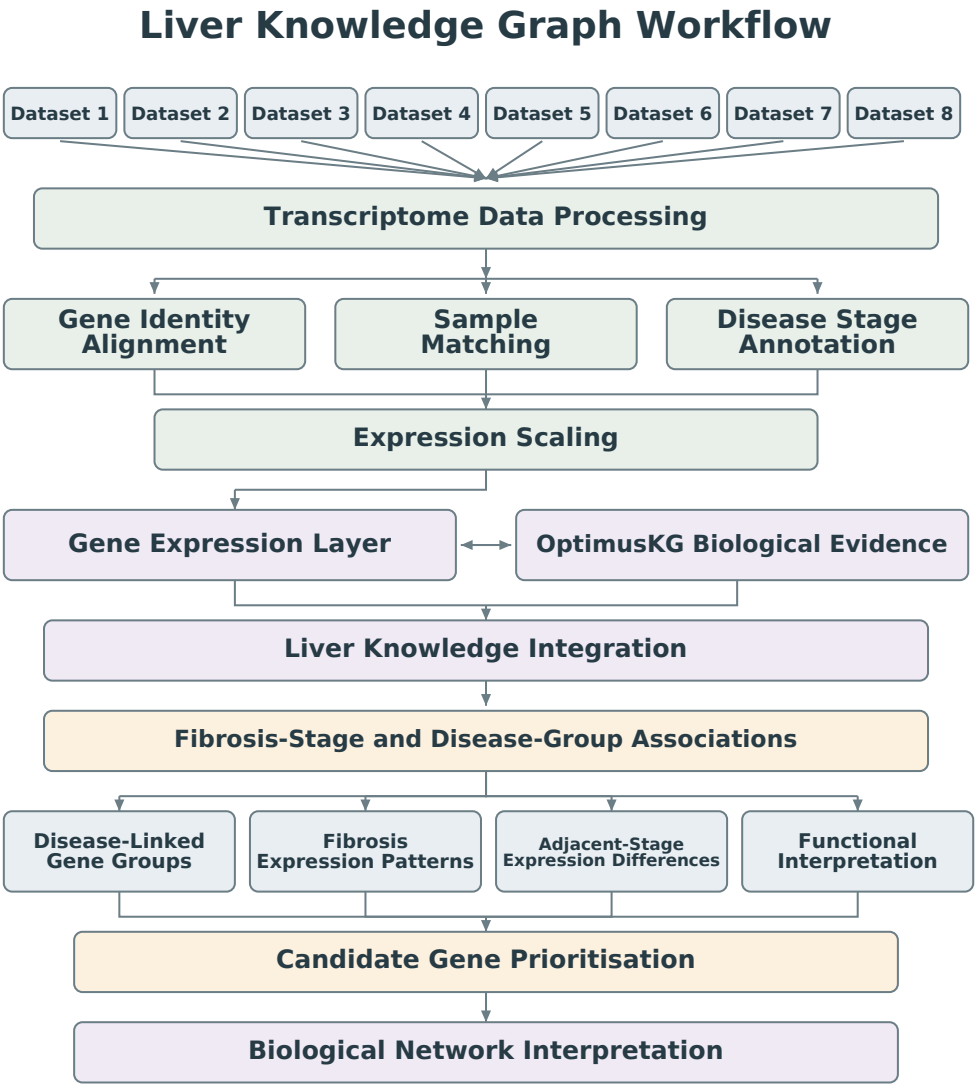


Figure 1. Study workflow. Eight public liver transcriptomic datasets undergo transcriptome processing, gene-identifier harmonisation, sample matching, clinical (disease-stage) annotation and within-study expression scaling to form the gene-expression layer, which is integrated with OptimusKG biological evidence into the liver knowledge graph. Cross-study fibrosis-stage and disease-group association analysis then informs disease-linked gene grouping, fibrosis-stage expression-pattern clustering, adjacent-stage expression comparisons and functional interpretation, followed by multi-evidence candidate-gene prioritisation and biological-network interpretation of the prioritised genes.

A common patient-level policy was applied to repeated biopsies in all analyses. When all recorded fibrosis stages of a patient were known and identical, standardised expression was averaged across biopsies; otherwise only the first biopsy was retained. Biopsy selection preceded filtering for individual stage comparisons, so a later biopsy could not enter a comparison solely because the first fell outside the stages compared. In GSE193066, 164 biopsies from 106 patients included 58 repeat pairs [28]: 28 had matching stages and were averaged, and 30 contributed only their first biopsy. Studies required at least 25 patient-level observations and variation in the relevant clinical variable. The fibrosis analysis comprised 610 patients from GSE130970, GSE135251, GSE162694, GSE193066 and GSE240729, and the disease-group analysis 216 patients from GSE126848, GSE130970 and GSE167523 (Additional file 1: Table S2).

Table 2. Public liver transcriptomic datasets and clinical annotation coverage. Counts are before repeat-biopsy resolution; recorded patient identifiers are counted within studies and do not establish distinct participants across accessions.

Accession	Ref.	Samples	Patient IDs	Staged biopsies	Disease labels[a]	Used in[b]
GSE126848	[27]	57	57	0	57	D
GSE130970	[10]	78	78	78	73[c]	F, D
GSE135251	[11]	216	216	216	216	F
GSE162694	[12]	143	143	143	143	F
GSE167523	[13]	98	98	0	98	D
GSE193066	[28]	164	106	164	0	F
GSE240729	[14]	67	67	67	0	F
GSE269412	[29]	262	262	0	0	Graph
Total		**1,085**	**1,027**	**668**	**587**	

[a]Includes broad NAFLD and obese-only labels excluded from the ordered analysis. [b]F, fibrosis-stage analysis (610 patients); D, ordered disease-group analysis (216 patients: 45 from GSE126848 after excluding 12 obese-only samples, 73 from GSE130970 and 98 from GSE167523); Graph, knowledge graph only. [c]Histology-derived labels; five ambiguous samples set to unclassified.

Gene harmonisation and expression preprocessing

Matrix, GCT and per-sample inputs were parsed into sample–gene measurements. Missing and non-numeric values were removed, measured zeros were retained and the first occurrence of duplicate raw identifiers was kept. The HGNC complete set was used to map approved symbols, Entrez identifiers and unambiguous previous symbols or aliases to Ensembl gene identifiers; version suffixes and `_PAR_Y` rows were removed, and measurements from distinct raw identifiers mapping to the same gene were summed. Count matrices were transformed to $\log_2(\mathrm{CPM}+1)$, with library sizes calculated before mapping and filtering; the GSE130970 TPM matrix to $\log_2(\mathrm{TPM}+1)$; and the deposited DESeq2 [30] relative-log-expression-normalised GSE193066 counts to $\log_2(\mathrm{count}+1)$ without additional rescaling. Each gene was then standardised within each study (mean 0, SD 1; $\mathrm{ddof}=1$; set to 0 when the SD was undefined), so effects are expressed in within-study SD units rather than fold changes; GSE193066 was standardised over all 164 biopsies before patient-level handling. No cross-study batch-correction model was fitted; instead, inference was performed within studies and combined across them [19]. The graph retained all 53,993 measured genes, whereas association testing used the 14,794 genes measured in all eight studies.

Cross-study association analysis

Within each eligible study, Spearman's ρ (average ranks for ties) was computed between patient-level standardised expression and fibrosis stage or disease group; rank correlation was used because both clinical axes are ordinal and expression changes need not be linear. Two-sided p-values were obtained from $t=\rho\sqrt{(n-2)/(1-\rho^2)}$ on $n-2$ degrees of freedom, with ρ clipped to ± 0.9999999. Genes testable in at least three studies were retained; the median study correlation summarised effect size and direction, and study p-values were combined by Fisher's method [31],

$$X_g^2 = -2\sum_{d=1}^{D_g} \ln p_{gd} \;\sim\; \chi^2_{2D_g}, \tag{1}$$

where D_g is the number of testable studies and p-values were bounded below at 10^{-300}. Benjamini–Hochberg (BH) correction [32] was applied separately to the fibrosis (14,794 genes) and disease-group (14,784 genes) families; ten genes with constant expression among the 45 eligible GSE126848 patients failed the three-study requirement on the latter axis. Because Fisher's statistic ignores direction, genes with $q < 0.05$ were additionally required to share the sign of the median correlation in every testable study, a conservative filter against associations driven by a single cohort [20]. No minimum effect size was imposed at this stage. Genes were labelled fibrosis-only, disease-group-only or shared (concordant or discordant); these labels describe selection outcomes across different study sets rather than biological specificity.

Fibrosis-stage expression profiles and clustering

For each fibrosis-associated gene, mean standardised expression was calculated at each stage and the five-point profile was standardised across stages ($\mathrm{ddof} = 0$), so that clustering reflected profile shape rather than magnitude. Profiles were clustered by k-means (50 initialisations; random seed 42). For $k = 3$–8 we computed the silhouette score [33] and the adjusted Rand index (ARI) [34] over 50 refits on random 80% gene subsets and over 50 refits resampling patients within studies (10 initialisations each); the largest k with mean gene-subset ARI ≥ 0.95 was selected (Additional file 1: Table S1). Membership confidence was estimated from 100 further patient bootstraps: bootstrap centroids were matched to reference centroids by Hungarian assignment [35] on centroid correlation, a gene was assessable in a bootstrap when the centroid matched to its reference cluster had $r \geq 0.5$, and confidence was the fraction of assessable bootstraps in which the gene returned to its cluster. Genes with confidence ≥ 0.8 formed the confidence core. Three sensitivity analyses were performed: profiles were re-estimated by ordinary least squares with categorical stage and study terms and reclustered; the $k = 3$ solution was recomputed with identical settings to assess nesting of the five patterns; and cluster assignments were compared with the transition at which each gene's study-adjusted coefficient was largest (next section).

Adjacent-stage comparisons and matched refits

For each transition (F0–F1, F1–F2, F2–F3, F3–F4), studies containing patients at both stages were modelled by ordinary least squares with study-specific intercepts:

$$z_{ig} = \alpha_g + \beta_g H_i + \sum_{d=2}^{D} \gamma_{gd}\, I(\mathrm{study}_i = d) + \varepsilon_{ig}, \tag{2}$$

where z_{ig} is the standardised expression of gene g in patient i and H_i indicates the higher stage. β_g was tested by two-sided t-test with BH correction across the fibrosis-associated genes within each contrast; differences with $q < 0.05$ and $|\beta_g| \geq 0.2$ were termed substantial. To assess sensitivity to unequal stage sizes, 30 patients per stage were drawn from the four studies represented at every stage (2 from GSE130970, 14 from GSE135251, 8 from GSE162694 and 6 from GSE240729), without replacement, in 50 draws (seed 42), and the models were refitted; reproducibility was the fraction of draws with $|\beta| \geq 0.2$ and the sign of the full-data coefficient. Study adjustment did not control for demographic, clinical or cell-composition differences.

Functional enrichment analysis

Over-representation was tested with GSEApy [36] (one-sided hypergeometric test) for MSigDB 2024.1.Hs GO Biological Process, KEGG legacy, Reactome and Hallmark collections [37, 38]. The background comprised the 14,794 genes in the common universe; gene sets were intersected with it and retained when they contained 10–500 background genes, and input groups required at least five genes. BH correction was applied within each group and collection ($q < 0.05$). Tests covered six concordant direction groups (fibrosis-only, disease-group-only and shared, each rising or falling; discordant shared genes excluded), the five clusters and their five confidence cores. Enrichment was interpreted as annotation over-representation, not measured pathway activity.

Gene prioritisation and robustness

Each fibrosis-associated gene received a score $S_g = P_g + T_g + X_g + F_g + B_g$ (maximum 9; full rules in Additional file 1: Table S3). P_g (association magnitude) awarded 2 points for $|\mathrm{median}\,\rho| \geq 0.40$ and 1 for $0.30 \leq |\mathrm{median}\,\rho| < 0.40$. T_g (endpoint separation) awarded 2 points for $|\bar{z}_{F4} - \bar{z}_{F0}| \geq 1$ and 1 for ≥ 0.5, provided that its sign agreed with the median correlation. X_g (transition evidence), for the transition with the largest absolute full-data coefficient, awarded one point for a substantial full-data difference and one for same-sign $|\beta| \geq 0.2$ in at least 50% of matched draws. F_g awarded one point for fibrosis-only selection. B_g (functional support) awarded 2 or 1 points when the gene belonged to a significant term in at least two or in one annotation collection, through its direction group or full cluster. Tier 1 required $S_g \geq 7$ and Tier 2 $5 \leq S_g < 7$; up to five Tier 1 genes per cluster were shortlisted by score and then absolute median correlation. Because the components share expression and annotation evidence, tiers are prioritisation categories rather than independent confirmations.

LOSO analysis omitted each of the five fibrosis studies in turn, repeated the patient-level association analysis and applied BH correction across all 14,794 genes; for Tier 1 genes we recorded retention of $q < 0.05$, direction and $|\mathrm{median}\,\rho| \geq 0.30$. Sensitivity analyses compared four label and normalisation configurations: original or unclassified labels for the five ambiguous GSE130970 samples, crossed with CPM-style rescaling or the deposited normalisation of GSE193066. The combined configuration (unclassified labels; deposited normalisation) was used for all primary analyses. We also examined threshold changes, removal of each score component and an alternative shortlist rule.

Knowledge-graph construction and annotation analysis

The expression layer and the biomedical knowledge layer were integrated in Neo4j by matching Ensembl identifiers to OptimusKG entities [25] (source deposit doi:10.7910/DVN/IYNGEV). The schema comprised Gene, Sample, Dataset, Disease, Drug, Pathway, BioProcess and Phenotype nodes and 15 relationship types. Sample–gene relationships stored raw and standardised measurements, and patient identity and biopsy order were stored as sample properties. The curated layer included Open Targets [39] gene–disease associations (score ≥ 0.1), deduplicated gene interactions, pathway and biological-process membership, disease hierarchy and phenotypes, and drug–gene and drug–disease relations, further enriched with drug annotations from ChEMBL, DGIdb, GtoPdb and Open Targets, KEGG and Hallmark pathways (MSigDB 2024.1), pathway–NAFLD links supported by both this study's enrichment results and CTD, and ChEMBL drug indications. These additions were not used in the analyses reported here (Additional file 1: Table S4).

Selected associations were stored with their direction, correlation, adjusted p-value and number of studies, and automated checks confirmed that all 8,044 progression links matched the analysis exports. For the annotation analysis, each Pathway or BioProcess node linked to at least one Tier 1 gene (3,980 nodes) was tested for over-representation of Tier 1 genes by hypergeometric test against the 14,794-gene background, with BH correction across all 3,980. Annotations with 10–500 background genes, at least ten Tier 1 genes and $q < 0.05$ were ranked by q, and an annotation was skipped if its Tier 1 members overlapped an already retained annotation with Jaccard similarity > 0.5.

Software

Analyses used Python 3.12.14 with pandas 2.3.3, NumPy 2.2.4, SciPy 1.16.3, scikit-learn 1.7.2 [40], matplotlib 3.10.7 and GSEApy 1.3.1, and Neo4j Enterprise 2026.09.0. Random seed 42 was used for clustering and matched refits. Reference-file versions and checksums are listed in Additional file 1.

Results

An integrated, patient-level resource of eight liver transcriptomic cohorts

The integrated resource comprised 1,085 samples and 1,027 recorded patient identifiers (Table 2). Five studies provided fibrosis stage for 668 biopsies, which yielded 610 patient-level observations after repeat-biopsy handling (F0–F4: 152, 160, 152, 112 and 34 patients), and three studies provided 216 observations with ordered disease-group labels. Gene harmonisation retained 53,993 measured genes, 14,794 of them measured in all eight studies. The integrated graph contained 126,999 nodes across eight entity types and 35,160,254 relationships across 15 types, including 32,471,042 expression measurements, 12,544 drug nodes, 25,890 drug–gene relationships and 8,044 gene–disease progression links (Additional file 1: Table S4).

Fibrosis and disease-group associations largely overlap

Cross-study testing identified 3,845 genes associated with fibrosis stage (of 14,794 tested) and 4,199 associated with disease group (of 14,784). The sets shared 1,822 genes and formed a union of 6,222, with 2,023 fibrosis-only and 2,377 disease-group-only genes (Fig. 2a). Fibrosis-only genes were balanced in direction (1,040 rising, 983 falling), whereas disease-group-only genes were predominantly rising (2,042 vs 335). Among shared genes, the median correlations on the two axes were strongly concordant (Spearman $\rho = 0.702$, $p = 1.9 \times 10^{-270}$): 1,408 genes rose and 384 fell on both axes, and only 30 were discordant (falling with fibrosis but rising with disease group; Fig. 2b). Individual associations were moderate, as expected for bulk tissue from heterogeneous cohorts (median $|\rho| = 0.226$, interquartile range 0.167–0.294; 890 genes with $|\rho| \geq 0.30$ and 165 with $|\rho| \geq 0.40$), so the analysis derives its power from consistency across cohorts rather than from large single-gene effects.

Fibrosis-associated genes follow five stage-dependent expression patterns

Clustering of the F0–F4 profiles of the 3,845 fibrosis-associated genes resolved five patterns (Fig. 3a,b; Table 3). C1 (857 genes) was higher at F1 than at F0 and higher again at each subsequent stage; C2 (1,367

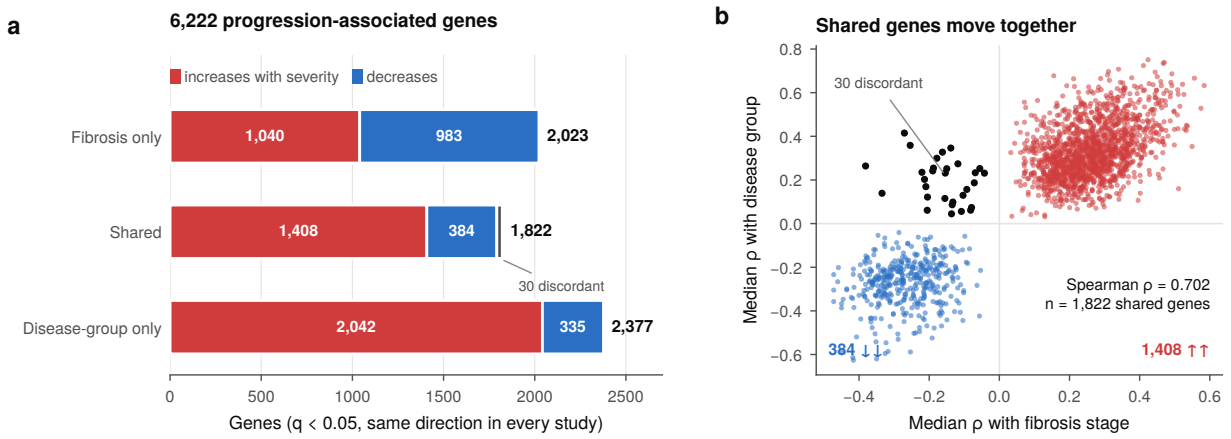


Figure 2. Cross-study gene associations. (a) Genes associated with fibrosis stage and/or ordered disease group (BH $q < 0.05$; same direction in every testable study; ≥ 3 studies), by direction. (b) Median within-study Spearman correlations of the 1,822 shared genes; red, concordant rising; blue, concordant falling; black, discordant. The disease-group axis orders source control, NAFL and NASH labels and is not a direct measure of inflammation.

Table 3. Fibrosis-stage expression clusters and their contribution to prioritisation. Mean F0 and F4 expression are in within-study SD units; core, membership confidence ≥ 0.8.

Cluster	Pattern	Genes	Core	Tier 1	Shortlist	Mean F0	Mean F4
C1	Higher from F1, increasing with stage	857	90	93	5	−0.295	0.515
C2	Increase, largest F3→F4	1,367	806	262	5	−0.227	0.860
C3	Higher from F1, no further increase	226	74	7	5	−0.270	0.136
C4	Decrease, largest F1→F2	397	126	3	3	0.161	−0.279
C5	Decrease, largest F3→F4	998	742	82	5	0.184	−0.658
Total		**3,845**	**1,838**	**447**	**23**		

genes), the largest cluster, increased mainly at higher stages, with its largest difference between F3 and F4; and C3 (226 genes) was higher at F1 than at F0 but did not increase further at later stages. Of the decreasing patterns, C4 (397 genes) decreased mainly between F1 and F2 and C5 (998 genes) mainly between F3 and F4. Because each patient contributes one stage, these profiles describe differences between patients at different stages rather than trajectories within individuals; throughout, "early" and "late" refer to lower and higher fibrosis stages, not to time.

The five patterns are robust to cluster number, modelling and resampling

The five-cluster solution was the largest k meeting the stability criterion (mean gene-subset ARI 0.983, SD 0.011), whereas patient resampling was more disruptive (ARI 0.592, SD 0.062) and the silhouette score was highest at $k = 3$ (0.587 vs 0.352 at $k = 5$; Additional file 1: Fig. S1a,b and Table S1). Because a reviewer could reasonably prefer the better-separated three-cluster solution, we tested how the two relate. The five patterns are refinements of the three-cluster solution rather than an alternative to it: re-running the clustering reproduced the published five-cluster labels exactly (ARI 1.000), and 91.9% of genes fell in the $k = 3$ cluster that contained most of their $k = 5$ cluster. C4 and C5 merged into a single falling group, C3 joined a group increasing at lower stages and C2 a group increasing at higher stages, and only C1 divided between the two rising groups (311 and 546 genes; Additional file 1: Fig. S1c). The five-cluster resolution therefore separates genes decreasing at intermediate stages from those decreasing between F3 and F4, and

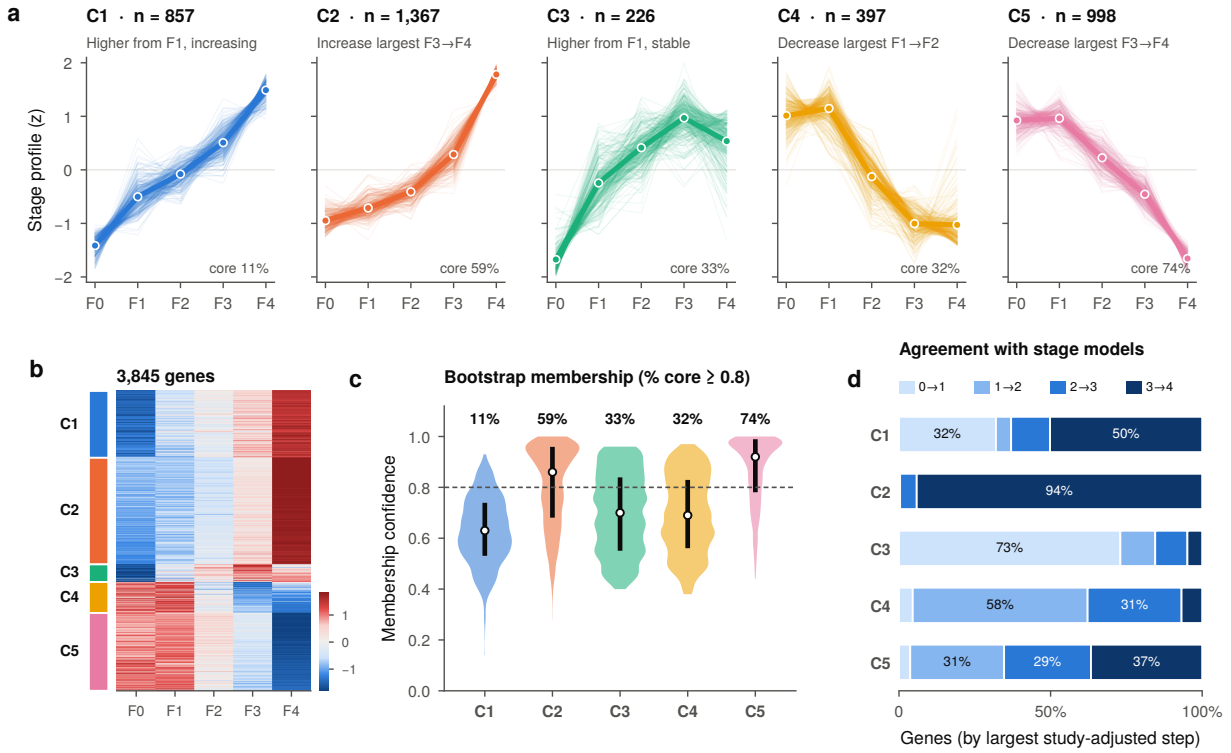


Figure 3. Five fibrosis-stage expression patterns and their robustness. (a) Gene profiles standardised across F0–F4: thin lines, individual genes (250 sampled per cluster); band, interquartile range; bold line, mean; "core", share of genes with membership confidence ≥ 0.8. (b) All 3,845 profiles grouped by cluster and ordered by membership confidence. (c) Membership confidence across 100 patient bootstraps (bar, interquartile range; dot, median). (d) Share of genes in each cluster by the adjacent-stage transition with the largest study-adjusted coefficient (Eq. 2). Profiles are cross-sectional stage associations, not longitudinal trajectories. Cluster-number diagnostics and nesting within the three-cluster solution are shown in Additional file 1: Fig. S1.

genes elevated from F1 without further increase from those increasing at higher stages, distinctions that the three-cluster solution conceals.

Cluster shapes also agreed with an independently specified model. For each gene, the adjacent-stage transition with the largest study-adjusted coefficient (Eq. 2) matched its cluster's defining transition: F3→F4 for 94% of C2 genes, F0→F1 for 73% of C3 genes and F1→F2 for 58% of C4 genes (Fig. 3d). Re-estimating profiles with study terms reproduced the clustering (ARI 0.901; median per-gene profile correlation 0.999). Average shapes were robust—median bootstrap centroid correlations ranged from 0.967 to 0.996—whereas individual assignments were less stable: 1,838 genes (48%) met the 0.8 confidence threshold, ranging from 11% of C1 to 74% of C5 (Fig. 3c). Confident genes had stronger stage associations (median $|\rho|$ 0.241 vs 0.214; Mann–Whitney $p = 2.9 \times 10^{-19}$), indicating that uncertain assignments arise mainly among weakly associated genes with intermediate profiles, such as the C1 genes that lie between the groups increasing at lower and at higher stages.

Expression differences concentrate at the transition to cirrhosis

Study-adjusted models identified substantial differences ($q < 0.05$ and $|\beta| \geq 0.2$) for 857, 1,048, 1,116 and 781 genes at the F0–F1, F1–F2, F2–F3 and F3–F4 transitions (Fig. 4b). Increasing/decreasing counts were 822/35, 459/589, 792/324 and 723/58: increases dominated the first and last transitions, whereas decreases predominated between F1 and F2, consistent with the timing of C4. The median absolute coefficient rose

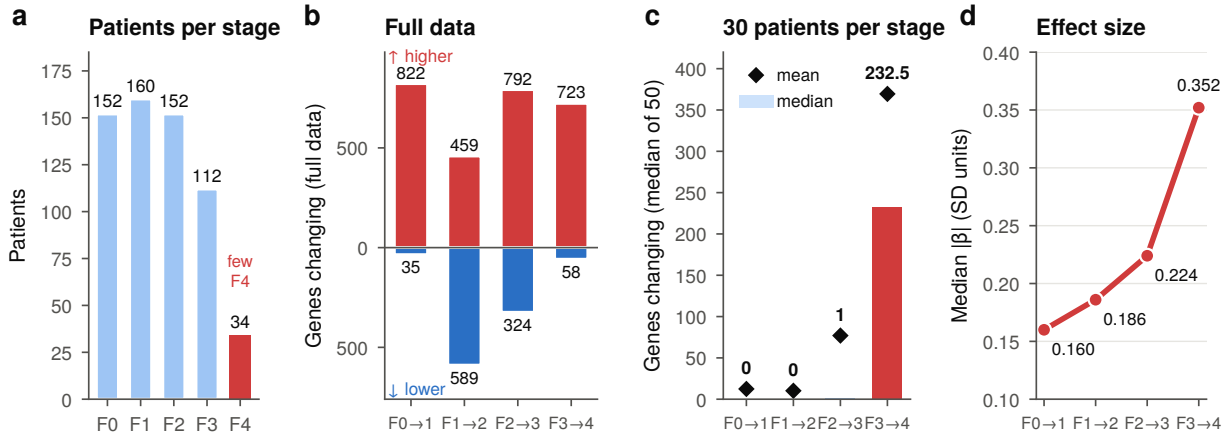


Figure 4. Adjacent-stage comparisons. (a) Patients per stage after repeat-biopsy handling. (b) Fibrosis-associated genes with substantial study-adjusted differences ($q < 0.05$, $|\beta| \geq 0.2$), higher (up) or lower (down) at the later stage. (c) Median (bars) and mean (diamonds) substantial-gene counts across 50 refits with 30 patients per stage and fixed study composition. (d) Median absolute study-adjusted coefficient (within-study SD units).

monotonically from 0.160 to 0.186, 0.224 and 0.352 (Fig. 4d). The first three comparisons included 152/160, 160/152 and 152/112 patients across five studies, and F3–F4 included 86/34 patients across four studies (Fig. 4a). In matched refits with 30 patients per stage, mean counts of substantial genes were 12.42, 10.22, 77.02 and 369.46 (SDs 67.6, 34.4, 154.2 and 383.4; medians 0, 0, 1 and 232.5; Fig. 4c). Thus, the large full-data counts for lower-stage transitions reflected their larger samples, whereas the F3–F4 transition remained prominent at equal power. This conclusion should be interpreted cautiously: the F3–F4 contrast rested on only 34 F4 patients from four studies, and because each matched draw required 30 of these 34 patients, successive draws shared most F4 observations, so the matched-refit distribution understates sampling uncertainty at F4. Estimates for the cirrhosis stage, including the F4 values of the C2 and C5 profiles, are therefore the least precise in the study.

Each pattern maps to a distinct functional programme

Functional enrichment distinguished immune, matrix-associated, proliferative and metabolic programmes (Fig. 5). Fibrosis-only rising genes were enriched for T-cell activation (80 overlapping genes; fold enrichment 2.281; $q = 8.19 \times 10^{-9}$) and Hallmark IL-6/JAK/STAT3 signalling (19 genes; 3.556; $q = 2.14 \times 10^{-5}$). Genes rising on both axes were enriched for matrix biology: collagen fibril organisation (GO Biological Process; 29 genes; 5.165; $q = 1.40 \times 10^{-11}$), collagen formation (Reactome; 32 genes; 4.151; $q = 1.47 \times 10^{-10}$) and extracellular-matrix organisation (Reactome; 74 genes; 2.956; $q = 3.80 \times 10^{-15}$). Among clusters, C2 was enriched for the Hallmark epithelial–mesenchymal transition set (74 genes; 4.215; $q = 5.25 \times 10^{-27}$), C3 for E2F targets (25 genes; 8.307; $q = 1.24 \times 10^{-14}$), and C5 for bile-acid (37 genes; 4.986; $q = 2.66 \times 10^{-15}$) and fatty-acid metabolism (36 genes; 3.606; $q = 1.42 \times 10^{-10}$). C1 and C2 had the broadest profiles (370 and 518 significant GO Biological Process terms; 18 and 20 Hallmark sets). Enrichment of C2 and C5 was robust to restricting the analysis to confidence-core genes (19 of 21 and all 5 core Hallmark sets also significant in the full clusters), whereas C4's single Reactome result changed from intrinsic fibrin-clot formation (full cluster; $q = 0.0114$) to BMP signalling (core; $q = 0.00150$), so its functional interpretation is tentative.

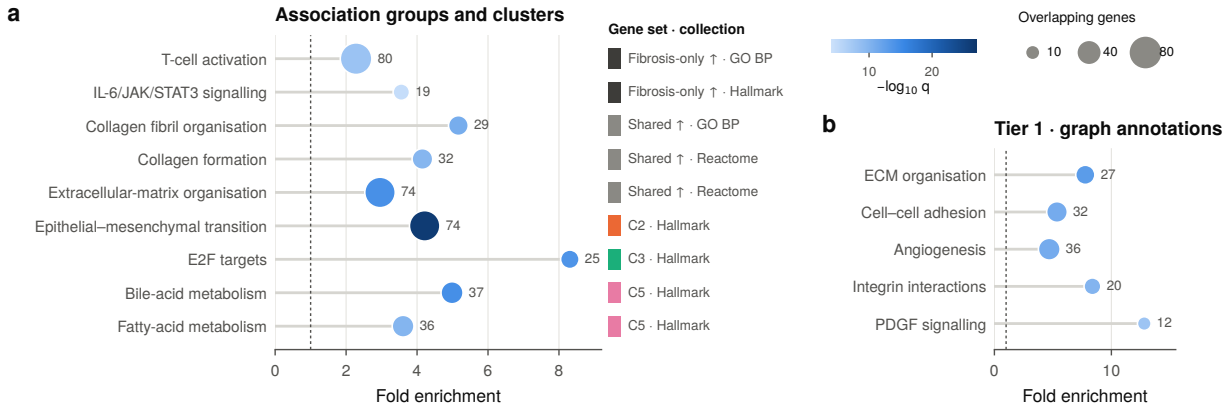


Figure 5. Functional enrichment and knowledge-graph context. (a) Representative gene sets over-represented in association groups and clusters (one-sided hypergeometric test against the 14,794-gene background; BH correction within each group and collection). (b) Knowledge-graph Pathway/BioProcess annotations over-represented among the 447 Tier 1 genes (BH correction across 3,980 annotations; redundant annotations removed). Position, fold enrichment; colour, $-\log_{10} q$; dot area and number, overlapping genes; dashed line, no enrichment.

Table 4. Cluster membership of established marker genes. ρ, median within-study correlation with fibrosis stage; conf., membership confidence; T1, Tier 1 (T2, Tier 2; T3, Tier 3).

Process	Cluster	Genes (ρ; conf.; tier)
Fibrillar matrix, stellate activation	C1	*COL1A1* (0.460; 0.78; T1), *COL1A2* (0.424; 0.60; T1), *COL3A1* (0.411; 0.77; T1), *COL5A1* (0.447; 0.56; T1), *PDGFRB* (0.313; 0.75; T1), *TGFB1* (0.338; 0.69; T1)
Basement membrane, remodelling	C2	*COL4A1* (0.388; 0.95; T1), *COL4A2* (0.400; 0.94; T1), *TIMP1* (0.440; 0.94; T1), *MMP2* (0.346; 0.93; T1), *LUM* (0.488; 0.68; T1), *VCAN* (0.464; 0.88; T1), *THBS2* (0.514; 0.79; T1), *PDGFRA* (0.417; 0.68; T1)
Ductular reaction	C2	*EPCAM* (0.365; 0.94; T1), *SOX9* (0.433; 0.91; T1), *KRT7* (0.341; 0.95; T1)
Macrophage, chemokine	C2/C3	*CCL2* (0.414; 0.97; T1; C2), *LGALS3* (0.427; 0.63; T1; C2), *TREM2* (0.469; 0.67; T1; C3)
Proliferation	C3	*CDC45* (0.401; 0.75; T1), *TYMS* (0.489; 0.61; T1), *TOP2A* (0.281; 0.86; T2), *MKI67* (0.305; 0.80; T2)
Hepatocyte metabolism	C5	*CYP2C19* (−0.443; 0.76; T1), *CYP2C8* (−0.310; 0.89; T1), *SULT1E1* (−0.414; 0.99; T1), *ACADSB* (−0.447; 0.89; T1), *CYP2E1* (−0.270; 0.99; T2), *CYP1A2* (−0.186; 0.95; T2), *ALB* (−0.128; 0.92; T2)

Canonical cell-type marker genes localise to specific patterns

To relate the patterns to the cellular events of fibrosis, we examined where established marker genes fell (Table 4). Fibrillar collagens (*COL1A1*, *COL1A2*, *COL3A1*, *COL5A1*), the stellate-cell receptor *PDGFRB* and *TGFB1* belonged to C1 (higher from F1 and at each later stage), and all reached Tier 1. Basement-membrane collagens (*COL4A1*, *COL4A2*), matrix-remodelling genes (*TIMP1*, *MMP2*, *LUM*, *DCN*, *VCAN*, *EFEMP1*), *THBS2*, *CCL2*, *LGALS3* and the ductular-reaction markers *EPCAM*, *SOX9* and *KRT7* belonged to C2, which increased most between F3 and F4, mostly with high membership confidence. Proliferation markers (*MKI67*, *TOP2A*, *CDC45*, *TYMS*) and *TREM2* belonged to C3, elevated from F1 without further increase, whereas cytochrome P450 genes (*CYP2E1*, *CYP1A2*, *CYP2C8*, *CYP2C19*), *SULT1E1*, *ACADSB* and *ALB* belonged to C5, which decreased most between F3 and F4.

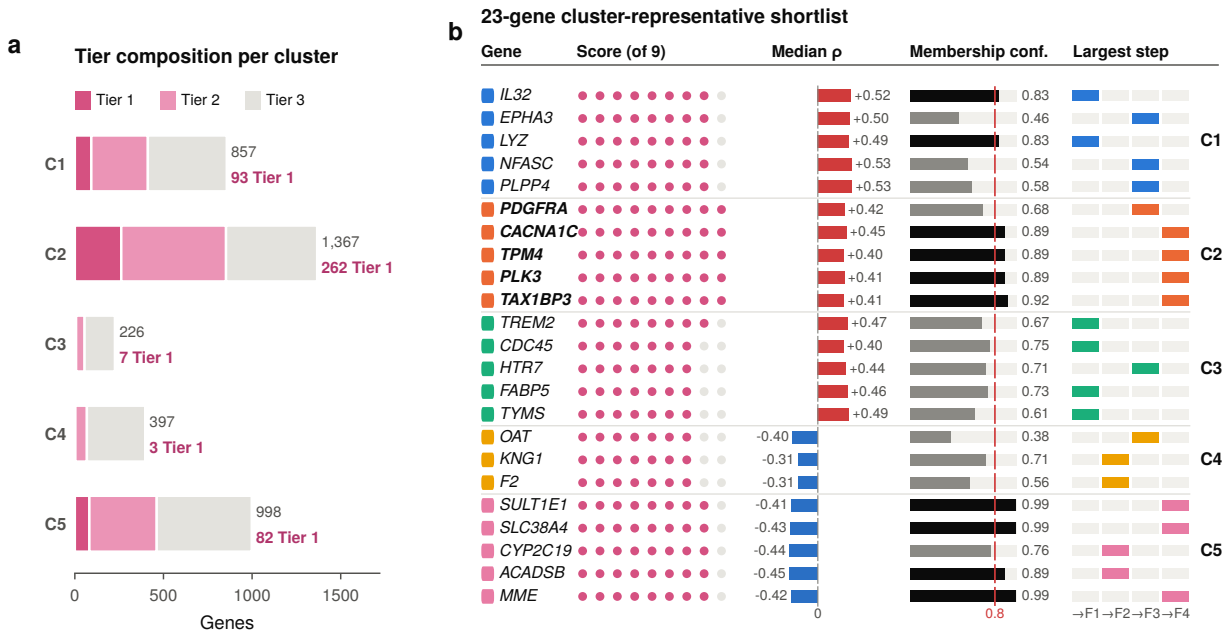


Figure 6. Multi-evidence gene prioritisation. (a) Tier composition per cluster (Tier 1, score ≥ 7; Tier 2, 5–6; Tier 3, < 5). (b) The 23-gene cluster-representative shortlist: total score (filled dots; bold, maximum score), median within-study correlation with fibrosis stage, membership confidence (black, ≥ 0.8; red line, core threshold) and the adjacent-stage transition with the largest study-adjusted coefficient, which need not coincide with the cluster's defining transition.

Multi-evidence prioritisation identifies 23 representative candidates

The multi-evidence score identified 447 Tier 1 genes: 93 in C1, 262 in C2, 7 in C3, 3 in C4 and 82 in C5 (Fig. 6a). Tier 1 genes had more reliable cluster assignments than other fibrosis-associated genes (71% vs 48% in confidence cores), although membership confidence was not part of the score. Selecting up to five Tier 1 genes per cluster produced a 23-gene shortlist representing all five patterns (Fig. 6b; Table 5); C4 contributed three candidates because only three of its genes reached Tier 1. *PDGFRA*, *CACNA1C*, *TPM4*, *PLK3* and *TAX1BP3* achieved the maximum score of nine. Membership confidence among shortlisted genes ranged from 0.38 (*OAT*) to 0.99 (*SULT1E1*, *SLC38A4*, *MME*), so a high score did not guarantee confident stage timing.

LOSO analysis retained significance for 442 of 447 Tier 1 genes in all five refits, whereas 247 retained $|\text{median } \rho| \geq 0.30$ throughout; all 23 shortlisted genes remained significant, and 18 retained the magnitude threshold. Significance retention was therefore more robust than effect-size stability. Sixteen shortlisted genes were selected under all four label and normalisation configurations, and removing the fibrosis-only component while keeping the threshold of seven yielded 286 Tier 1 genes and a 21-gene shortlist (with a lower maximum achievable score).

Knowledge-graph annotations converge on matrix, adhesion and PDGF signalling

The enrichment-ranked subgraph connected 42 Tier 1 genes to 12 non-redundant Pathway/BioProcess annotations through 154 relationships (Additional file 1: Fig. S2). Tier 1 genes were over-represented in extracellular-matrix organisation (27 genes; fold enrichment 7.77; $q = 1.06 \times 10^{-13}$), cell–cell adhesion (32 genes; 5.35; $q = 8.60 \times 10^{-12}$), angiogenesis (36 genes; 4.69; $q = 8.60 \times 10^{-12}$), integrin cell-surface

Table 5. Final 23-gene shortlist. Biological functions and prior liver-disease evidence for each gene are summarised in Additional file 1: Table S6. Median ρ, median within-study Spearman correlation with fibrosis stage; valid bootstraps, denominator for membership confidence; largest contrast, adjacent-stage comparison with the largest absolute study-adjusted coefficient.

Gene	Cluster	Score /9	Median ρ	Membership conf.	Valid bootstraps	Largest contrast
IL32	C1	8	0.522	0.83	100	F0→F1
EPHA3	C1	8	0.498	0.46	100	F2→F3
LYZ	C1	8	0.487	0.83	100	F0→F1
NFASC	C1	8	0.532	0.54	100	F2→F3
PLPP4	C1	8	0.533	0.58	100	F2→F3
PDGFRA	C2	9	0.417	0.68	97	F2→F3
CACNA1C	C2	9	0.447	0.89	97	F3→F4
TPM4	C2	9	0.403	0.89	97	F3→F4
PLK3	C2	9	0.414	0.89	97	F3→F4
TAX1BP3	C2	9	0.408	0.92	97	F3→F4
TREM2	C3	8	0.469	0.67	99	F0→F1
CDC45	C3	7	0.401	0.75	99	F0→F1
HTR7	C3	7	0.440	0.71	99	F2→F3
FABP5	C3	7	0.464	0.73	99	F0→F1
TYMS	C3	7	0.489	0.61	99	F0→F1
OAT	C4	7	−0.398	0.38	100	F2→F3
KNG1	C4	7	−0.312	0.71	100	F1→F2
F2	C4	7	−0.307	0.56	100	F1→F2
SULT1E1	C5	8	−0.414	0.99	96	F3→F4
SLC38A4	C5	8	−0.430	0.99	96	F3→F4
CYP2C19	C5	8	−0.443	0.76	96	F1→F2
ACADSB	C5	8	−0.447	0.89	96	F1→F2
MME	C5	8	−0.421	0.99	96	F3→F4

interactions (20 genes; 8.38; $q = 1.08 \times 10^{-10}$) and signalling by PDGF (12 genes; 12.81; $q = 1.28 \times 10^{-8}$) (Fig. 5b). Four shortlisted genes appeared in the subgraph: *PDGFRA* linked to five annotations and *EPHA3*, *NFASC* and *TREM2* to one each. The remaining displayed genes came from the wider Tier 1 set, so the network illustrates annotation sharing within a subset of prioritised genes; these curated links provide biological context and do not establish causal mechanisms.

Discussion

This study shows that liver fibrosis severity is associated with distinct stage-dependent transcriptional programmes rather than a single monotonic expression response. By integrating eight cohorts at the patient level and requiring the same direction of association in every study, we identified 3,845 fibrosis-associated genes and resolved their F0–F4 profiles into five patterns with different timing, functional content and cellular correlates. Three features distinguish these results from earlier grouped-stage analyses: the patterns were reproduced by study-adjusted stage models fitted directly to patient-level data; they nested coherently within the coarser three-cluster solution; and the transition to cirrhosis was the only step that retained many substantial differences at equal sample size, although this transition was also the one represented by the fewest patients (34 at F4). Together, they indicate that higher fibrosis stages are associated with a fibrogenic matrix programme already elevated at F1, a proliferative signature elevated from F1 without further increase,

and remodelling and metabolic differences that, in the available data, are most pronounced between F3 and F4.

The fibrillar matrix programme was already elevated at F1 and was higher at each subsequent stage. Fibrillar collagens (*COL1A1*, *COL1A2*, *COL3A1*, *COL5A1*), *PDGFRB* and *TGFB1* belonged to C1, which already differed between F0 and F1 and was higher at each subsequent stage up to F4. Lineage-tracing studies have identified activated hepatic stellate cells as the dominant source of fibrillar collagen across aetiologies [41], with TGF-β and PDGF signalling as principal drivers of their activation [42]. The elevation of this cluster at F1 therefore suggests that the fibrogenic programme of stellate-cell activation is already detectable transcriptionally in patients at F1, before histologically significant ($\geq$F2) fibrosis. In parallel, fibrosis-only rising genes were enriched for T-cell activation and IL-6/JAK/STAT3 signalling, pathways that regulate stellate-cell activation and hepatocyte survival in fibrosis [43]. Because these genes were selected on the fibrosis axis but not on the disease-group axis, they are candidates for immune processes that track scar formation more closely than steatohepatitis activity, although the different cohorts contributing to each axis prevent a claim of specificity.

A second, distinct programme was elevated from F1 onward but did not increase further at higher stages. C3 was enriched for E2F targets and contained proliferation markers (*MKI67*, *TOP2A*, *CDC45*, *TYMS*) together with *TREM2*. A higher proliferative signal at mild stages is consistent with regenerative and ductular responses to injury, and the absence of further increase at higher stages would be compatible with the replicative arrest and senescence of hepatocytes reported with advancing NAFLD [44], although bulk data cannot identify which cells proliferate. *TREM2* marks the NASH-associated macrophages that expand in experimental steatohepatitis [45] and the scar-associated macrophages that populate the fibrotic niche in cirrhosis [46]. Its higher expression from F1 without further increase at higher stages may indicate that this macrophage state is already present in patients with mild fibrosis; because the data are cross-sectional, longitudinal and cell-resolved studies are needed to test this.

The C2 programme, the largest cluster and the strongest functional signal in the study, changed most steeply between F3 and F4; because F4 was represented by 34 patients, the magnitude and stage location of this change should be regarded as provisional. Its Hallmark epithelial–mesenchymal transition enrichment should not be read as evidence of hepatocyte EMT, which lineage tracing has not supported [47]. Instead, the cluster's composition—basement-membrane collagens, *TIMP1*, *MMP2*, proteoglycans, *PDGFRA* and the ductular-reaction markers *EPCAM*, *SOX9* and *KRT7*—points to mesenchymal expansion, vascular and basement-membrane remodelling and ductular reaction, all of which increase with fibrosis severity in human NAFLD [48]. Knowledge-graph analysis converged on the same biology: Tier 1 genes were over-represented in extracellular-matrix organisation, cell adhesion, angiogenesis, integrin interactions and PDGF signalling. Integrin-mediated activation of latent TGF-β in PDGFRβ^+ mesenchymal cells is a core fibrogenic mechanism across organs [49], and PDGFRα^+ scar-associated mesenchymal cells expand in human cirrhosis [46]. *PDGFRA*, which scored maximally and was the shortlisted gene most connected to these annotations, is therefore a strong candidate for cell-resolved follow-up. Notably, *THBS2* was among the most strongly fibrosis-associated C2 genes ($\rho = 0.514$); thrombospondin-2 was independently identified as a circulating biomarker of NASH and advanced fibrosis [13], illustrating how secreted C2 products that increase at advanced stages could be evaluated as non-invasive markers of advanced disease.

The falling programmes describe the loss of hepatocyte function. C5, which fell most steeply between F3

and F4, was enriched for bile-acid and fatty-acid metabolism and contained drug-metabolising enzymes (*CYP2E1*, *CYP1A2*, *CYP2C8*, *CYP2C19*), *SULT1E1* and *ALB*. Altered hepatic cytochrome P450 expression and activity across the stages of NAFLD have been documented directly [50]; our results place the decline of these genes predominantly at the advanced stages, which is relevant to drug metabolism in patients with advanced fibrosis. In bulk tissue, such decreases may reflect reduced hepatocyte content as much as altered regulation; deconvolution approaches [51] or single-cell data would be required to separate the two. The mid-stage falling cluster C4 was small, and its enrichment did not replicate in confidence-core genes, so we refrain from interpreting it beyond noting its coagulation-related genes (*F2*, *KNG1*).

Although no independent cohort was analysed here, several key findings agree with evidence generated independently of our fibrosis analysis. *TREM2*$^+$ macrophages and PDGFRα^+ mesenchymal cells were identified in human cirrhotic liver by single-cell sequencing [46]; thrombospondin-2, encoded by the C2 gene *THBS2*, was identified as a circulating biomarker of NASH and advanced fibrosis [13] in a cohort (GSE167523) that contributed no fibrosis stages to our fibrosis-axis analysis; and stage-dependent alterations of hepatic cytochrome P450 enzymes, matching the C5 genes, were measured directly at the protein and activity level [50]. These concordances support, but do not substitute for, replication of the stage patterns in independent cohorts.

These findings complement previous transcriptomic studies. The matrix and collagen programmes that dominated our shared rising genes agree with the fibrosis signatures of Govaere *et al.* and Hoang *et al.* [11, 10] and with the fibrogenic programmes shared across aetiologies [17]. Comparisons of advanced with mild fibrosis [18, 16] necessarily pool C1 and C2 genes, whose largest changes occur at different stages; separating them shows that the fibrillar matrix programme already differs at mild stages, whereas basement-membrane remodelling, ductular reaction and metabolic loss differ most between F3 and F4. The strong concordance between the fibrosis and disease-group axes (1,792 of 1,822 shared genes) further indicates that much of the transcriptional response is common to steatohepatitis and fibrosis.

The framework also has methodological implications. Combining heterogeneous public cohorts without a fitted batch-correction model relied on within-study standardisation, rank-based tests that respect the ordinal stage scale and a requirement that every testable study agree in direction, an approach that limits the influence of study-specific artefacts [21, 19, 20]. Resolving repeat biopsies before stage filtering prevented over-weighting of patients with several biopsies, and size-matched refits separated biological from power-driven differences between transitions. The cluster analysis illustrates why stability should be reported at two levels: average shapes were highly reproducible, but fewer than half of the genes had confident assignments, and confident genes were those with stronger stage associations. Reporting membership confidence, LOSO retention and preprocessing sensitivity alongside each gene makes the strength of each type of evidence explicit, and integrating these results in a knowledge graph keeps every candidate traceable from its expression in individual biopsies to its curated pathway, disease and drug relationships [22, 39].

The stage-resolved view has implications for translational work. Genes that already differ between F0 and F1 (C1 and C3) are candidate markers of mild fibrosis, whereas C2 and C5 genes, which differ most between F3 and F4, are more likely to mark advanced disease. Because individual early differences were small at equal sample size, early staging may require multi-gene scores rather than single markers. The prioritised genes, each annotated with its pattern, confidence and robustness, provide a ranked starting point for such studies, and the 16 shortlisted genes stable across all preprocessing variants are the most defensible

candidates.

This study has limitations. First, the cohorts were identified by a targeted rather than a formal systematic search, so other eligible datasets may exist. Second, participant overlap across accessions cannot be fully excluded from metadata; overlap between two cohorts would make them non-independent, inflating combined p-values and directional agreement, although LOSO retention limits the dependence of any result on a single cohort. Third, clinical confounding remains possible: age and sex were available for only 327 of the 610 staged patients, metabolic and alcohol data were incomplete, which also prevented reclassification under MASLD/MASH criteria [4], and study adjustment did not account for clinical covariates or cell composition; disease-group labels for GSE130970 were histology-derived proxies. Fourth, the cirrhosis stage was represented by only 34 patients, so conclusions about the F3–F4 transition and F4 expression levels are the least certain. All analyses were cross-sectional, so stage patterns describe differences between patients rather than trajectories within them. Finally, the score components share expression and annotation evidence, candidate selection was not nested within the robustness analyses, and independent, longitudinal and functional validation remain to be performed. The findings should therefore be regarded as prioritised hypotheses and a structured resource for their evaluation.

Conclusions

Liver fibrosis severity is associated with at least five distinct expression patterns across fibrosis stages: a fibrogenic matrix programme elevated from F1 and higher at each later stage, a proliferative signature elevated from F1 without further increase, and basement-membrane remodelling, ductular reaction and metabolic loss that are most pronounced between F3 and F4, a transition represented by few patients and therefore estimated least precisely. These patterns were robust to cluster number, modelling and resampling, and their marker genes are consistent with established cellular and biological processes associated with the fibrotic liver. The integrated, uncertainty-aware resource allows each candidate gene to be examined together with the evidence for its ranking and its stage assignment. Three priorities follow: testing the stage patterns and prioritised genes in independent and longitudinal cohorts with harmonised clinical labels and covariates, with larger numbers of patients with cirrhosis; using single-cell or spatial data to separate changes in cell state from changes in cell composition, particularly for *PDGFRA*, *TREM2* and the C2 ductular programme; and public release of the code, graph exports and complete outputs to support independent reproduction.

Abbreviations

ARI: adjusted Rand index; BH: Benjamini–Hochberg; CPM: counts per million; EMT: epithelial–mesenchymal transition; GEO: Gene Expression Omnibus; GO: Gene Ontology; HGNC: HUGO Gene Nomenclature Committee; LOSO: leave-one-study-out; MASH: metabolic dysfunction-associated steatohepatitis; MASLD: metabolic dysfunction-associated steatotic liver disease; NAFL: non-alcoholic fatty liver; NAFLD: non-alcoholic fatty liver disease; NASH: non-alcoholic steatohepatitis; SD: standard deviation; TPM: transcripts per million.

Supplementary Information

Additional file 1: Supplementary Figs. S1–S2, Tables S1–S6 and Supplementary Methods (cluster-number diagnostics and nesting analysis, biological context of the 23 shortlisted genes, knowledge-graph subgraph, dataset-specific preprocessing and eligibility, scoring rules, graph composition, expression files, implementation checks and reference-file provenance).

Declarations

Ethics approval and consent to participate. Not applicable. This study analysed only publicly available, de-identified data deposited in GEO under the original studies' ethical approvals; no participants were recruited and no samples were collected.

Consent for publication. Not applicable.

Availability of data and materials. The transcriptomic datasets are available from GEO under accessions GSE126848, GSE130970, GSE135251, GSE162694, GSE167523, GSE193066, GSE240729 and GSE269412. The OptimusKG source deposit is available at doi:10.7910/DVN/IYNGEV. Drug annotations were obtained from ChEMBL (REST API, October 2026), DGIdb 5, GtoPdb 2026.3.1 and Open Targets Platform 26.09. Analysis code, graph exports and complete result tables are available at `https://github.com/nissankararaoThanvitha/liver_graph` and archived at [Zenodo DOI].

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. []

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